supply-chain due diligence, and Digital Product Passports, depending on the product and market.

These requirements reach into engineering and supplier management. A material selected during design can determine chemical compliance; supplier declarations can determine whether compliance can be demonstrated; and substance, component and circularity data can trigger reporting or market-access requirements.

Compliance therefore cannot be treated as paperwork after manufacturing. For globally ambitious Indian electronics companies, material, component, supplier and regulatory-data systems increasingly need to be built into product development from the beginning.

Sustainability starts on the drawing board

Growth also creates resource and end-of-life challenges. More electronics means greater demand for materials and energy, while major markets increasingly regulate products across their lifecycle.

Engineers are consequently being asked whether devices can be repaired, components replaced, materials recovered, hazardous substances controlled, and recycled content incorporated without compromising reliability. Product information must also support reuse and recycling.

These are increasingly engineering questions. Because India's manufacturing capacity and product-development systems are still expanding, newer facilities can integrate quality, environmental compliance, traceability, and circularity early rather than retrofitting them later.

People create technological depth

Factories create capacity; people turn capacity into capability. A deeper electronics ecosystem needs semiconductor and system designers, process and equipment engineers, packaging and photonics specialists, materials scientists, embedded-system developers, reliability experts and advanced-manufacturing technicians, alongside quality, regulatory, data and supply-chain professionals.

The next objective is not simply more jobs, but greater technological depth: experienced engineers, specialist suppliers, research institutions, design companies, testing laboratories and accumulated manufacturing knowledge.

MSMEs: small firms, big difference

Large fabs and smartphone factories attract headlines, but mature electronics ecosystems depend on specialised smaller companies. Indian Micro, Small, and Medium Enterprises (MSMEs) have opportunities in components, precision manufacturing, tooling, equipment, materials, testing, embedded software, product design and engineering services.

Their integration into international supply chains will be an important measure of progress. At the Department of Commerce's Chintan Shivir on July 10, 2026, government and industry representatives discussed a roadmap towards US$150 billion in electronics exports by 2030, alongside stronger global value-chain participation and supply-chain resilience.

Reaching that ambition requires competitiveness across technology, cost, quality, reliability, compliance, logistics, and innovation. No single incentive can provide all of these. They require an ecosystem.

FROM SCALE TO TECHNOLOGICAL DEPTH

Growing Strength Today	Opportunity For Tomorrow
Final-product manufacturing	Deeper component manufacturing
Mobile-phone scale	Broader electronics categories
Assembly	Higher domestic value addition
Component localisation	Critical-material resilience
Semiconductor investment	Complete semiconductor ecosystem
Chip production	Advanced packaging and integration
Engineering talent	Indian design and intellectual property
Export growth	Deeper global value-chain participation
Manufacturing scale	World-class quality and reliability
Regulatory compliance	Compliance by design
E-waste management	Circular product design

From scale to technological depth

India's next electronics transformation can be viewed as a set of transitions as shown in the table above.

Together, these shifts offer a more meaningful definition of electronics self-reliance. It is not simply the absence of imports; it is the presence of capabilities.

What should India@100 measure?

As India marked its 80th Independence Day on August 15, 2026, there was much to celebrate. Electronics production has multiplied, exports have accelerated, mobile manufacturing has transformed, component localisation is gathering momentum, and semiconductor projects have moved from approvals into commercial production.

Yet scale alone will not define the next phase. By India@100, progress should also be measured by the chips India designs, the components and critical technologies it develops, the intellectual property it owns, the engineers and enterprises it nurtures, the resilience of its supply chains, and the quality and reliability Indian products demonstrate in global markets. The journey from 'Made in India' to 'Designed and engineered in India, trusted worldwide' may ultimately provide the clearest measure of technological self-reliance. EFY